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Batch: A-2
Division: A

Subject: Programming with Java

ASSIGNMENT No : 11

TITLE : Write a Java program to demonstrate the implementation of interfaces.

Problem Statement

Write a Java program to demonstrate the implementation of interfaces.

Learning Objectives

To implement interfaces for achieving abstraction and multiple inheritance of type.

Theory

An interface specifies abstract methods that implementing classes must define. It provides abstraction and supports multiple inheritance of type.

Output Screenshots

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp11/Main.java
Dog barks.
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

Exercise

Q1. Create an interface Printable and implement it in Student and Employee classes.

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp11/Exercise/Ex1.java
Student Name : Devansh
Roll No      : 101

Employee Name : Rahul
Employee ID   : 501
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

Q2. Create an interface Switchable with a method turnOn(). Implement the interface in Light and Fan classes to display the device status.

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp11/Exercise/Ex2.java
Light is ON
Fan is ON
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

Conclusion

This assignment helped in understanding the concept and implementation of interfaces in Java. It demonstrated how interfaces provide complete abstraction by defining a common set of methods that different classes can implement according to their specific requirements. The implementation improved my understanding of object-oriented programming concepts such as abstraction, polymorphism, and loose coupling, resulting in flexible and reusable code. These concepts are widely used in developing desktop applications, web applications, Android applications, enterprise software, banking systems, and other real-world Java applications that require modular, scalable, and maintainable software design.